

5. Implement a java program to illustrate the use of different types of character extraction, string comparison, string search and string modification methods.

```
import java.lang.*;
public class StringMethodsDemo {
    public static void main(String[] args) {

        String s1 = "Java Programming";
        String s2 = "Java";

        //Character Extraction
        System.out.println("Character at index 2: " + s1.charAt(2));
        System.out.println("Substring: " + s1.substring(5,16));

        //String Comparison
        System.out.println("Equals: " + s1.equals(s2));
        System.out.println("CompareTo: " + s1.compareTo(s2));

        //String Search
        System.out.println("Index of 'Pro': " + s1.indexOf("Pro"));
        System.out.println("Last index of 'a': " + s1.lastIndexOf('a'));

        //String Modification
        System.out.println("Replace: " + s1.replace("Java","Advanced Java"));
        System.out.println("Upper Case: " + s1.toUpperCase());
    }
}
```

Character at index 2: v Substring: Programming Equals: false CompareTo: 11 Index of 'Pro': 5 Last index of 'a': 3 Replace: Advanced Java Programming
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6. Implement a java program to illustrate the use of different types of StringBuffer methods

```
import java.lang.*;

public class StringBufferDemo {
    public static void main(String[] args) {

        StringBuffer sb = new StringBuffer("Java");

        // Append method
        sb.append(" Programming");
        System.out.println("Append: " + sb);

        // Insert method
        sb.insert(5, "Advanced ");
        System.out.println("Insert: " + sb);

        // Replace method
        sb.replace(5, 13, "Core ");
        System.out.println("Replace: " + sb);

        // Delete method
        sb.delete(5, 10);
        System.out.println("Delete: " + sb);

        // Reverse method
        sb.reverse();
        System.out.println("Reverse: " + sb);
    }
}
```

Append: Java Programming

Insert: Java Advanced Programming

Replace: Java Core Programming

Delete: Java Programming

Reverse: animmardorP avaJ